



Solar System Components

Panels · Inverters · Batteries · Controllers · BOS

• SOLAR ENERGY • LESSON 03

The Five Core Solar System Components

A solar PV system is not just panels on a roof. It is a complete electrical system with five major subsystems that must be correctly sized, matched, and wired to deliver reliable, safe power. Whether the system is a 2 kW residential off-grid installation or a 500 kW commercial on-grid array, the same five components are always present.

Component	Function	Key Variants
Solar Panels (PV Modules)	Convert sunlight to DC electricity via the photovoltaic effect	Monocrystalline, polycrystalline, thin-film, bifacial
Inverter	Convert DC power from panels to usable AC power (or manage DC battery)	String, microinverter, hybrid (battery-ready), off-grid
Battery Storage	Store excess solar energy for use when panels are not producing	Lead-acid (FLA/AGM/Gel), Lithium-ion (LFP, NMC)
Charge Controller	Regulate current from panels to batteries; prevent overcharge	PWM (Pulse Width Modulation), MPPT (Maximum Power Point Tracking)
Balance of System (BOS)	All other electrical and mechanical components that complete the system	Combiner box, disconnects, breakers, energy meter, racking, wiring, SPD

Complete Solar System Flow



Figure 1 — Complete Solar PV System Block Diagram

Solar Panels — The Power Source

Technology	Efficiency	Cost	Best For
Monocrystalline (Mono-Si)	19–24%	High	Limited roof space; maximum output per m²; most common residential
Polycrystalline (Poly-Si)	15–18%	Medium	Large ground-mount arrays; budget installations
Thin-Film (CdTe / CIGS)	10–13%	Low	Commercial flat roofs; building-integrated PV; partial shade tolerance
Bifacial Monocrystalline	20–26% (front+rear)	High	Ground mount with reflective surface; vertical installations; carports
Half-Cut / Multi-Busbar	20–23%	Medium-High	Shade-tolerant upgrade on mono cells; reduced resistive losses

Key panel spec: Always check the STC rating (Standard Test Conditions: 1 000 W/m², 25°C, AM1.5). Real-world output is typically 75–85% of STC due to temperature derating (panels lose ~0.35–0.45%/°C above 25°C) and system losses.

Inverters — The Brain of the System

The inverter is the most critical and most complex component. It performs **Maximum Power Point Tracking (MPPT)** — continuously adjusting the operating point of the panels to extract maximum power — and converts that DC energy into grid-compatible 230 V / 50 Hz AC power.

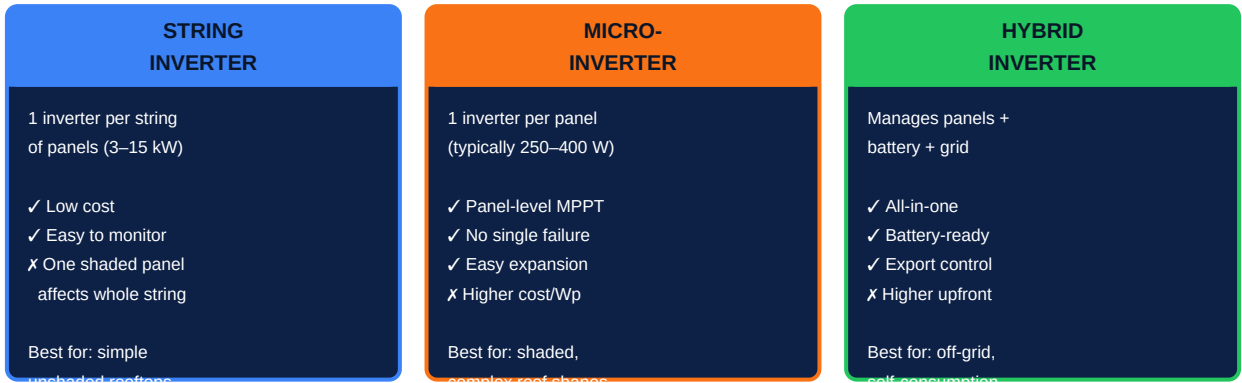


Figure 2 — Inverter Types: String vs Micro vs Hybrid

Inverter Spec	What It Means	Typical Range
Rated Output Power (kW)	Continuous AC power the inverter can deliver at full load	1.5–100+ kW residential/commercial
MPPT Channels	Number of independent DC string inputs with separate MPPT trackers	1–6 channels; more = better shade tolerance
MPPT Voltage Range	Min–max DC input voltage the inverter can track efficiently	200–800 V DC typical; higher = fewer series losses
Efficiency (European / CEC)	Weighted efficiency across typical operating conditions, not peak	96–99% for modern string inverters
IP Rating	Ingress protection against dust and water. IP65 min for outdoor	IP65 (outdoor wall), IP54 (indoor/sheltered)
Warranty	Most string inverters: 5 years standard, extendable to 10–25 years	Microinverters typically 25-year warranty

Charge Controllers — Off-Grid Essential

In off-grid and hybrid systems, a charge controller sits between the panels and the battery bank. It prevents overcharging (which destroys batteries) and manages the charge stages: bulk, absorption, and float.

Type	How It Works	Efficiency	When to Use
PWM (Pulse Width Modulation)	Directly connects panels to battery. Reduces voltage to match battery by switching rapidly.	~75%	Small systems (<200 W); panels matched to battery voltage; low-budget installs
MPPT (Max Power Point Tracking)	DC-DC converter steps down panel voltage to battery voltage, capturing extra power.	93–98%	All serious off-grid systems; high-voltage panel strings; cold climates

MPPT advantage: If your panels run at 72 V and your battery is 24 V, a PWM controller wastes the voltage difference. An MPPT controller converts that extra voltage into current — gaining up to 30% more energy in cold mornings when panel voltage is highest.

Battery Storage and Balance of System

Batteries allow a solar system to serve loads at night or during grid outages. The choice of battery chemistry fundamentally affects system cost, lifespan, maintenance requirements, and depth of discharge.

Battery Chemistry Comparison

Chemistry	DoD	Cycle Life	Cost/kWh	Notes
Flooded Lead-Acid (FLA)	50%	500–800	Low	Cheapest upfront; needs ventilation & regular watering; heavy
AGM (Sealed Lead-Acid)	50%	500–700	Low-Medium	Maintenance-free; no ventilation needed; heavier than Li-ion
Gel Lead-Acid	50%	500–1 000	Medium	Good deep-cycle; sensitive to overcharge; good in heat
LFP — Lithium Iron Phosphate	80–90 %	3 000–6 000+	High	Safest Li chemistry; long life; dominant in solar storage 2024+
NMC — Lithium Nickel Manganese Cobalt	80%	1 500–3 000	High	Higher energy density; used in EV-based home batteries (e.g. Powerwall)

Balance of System (BOS) Components



Figure 3 — Balance of System (BOS) Components

BOS Item	Purpose	Sizing Rule
DC Combiner Box	Joins multiple panel strings into one DC feed; contains fuses per string	1 fuse per string at $1.56 \times I_{sc}$; rated $\geq V_{oc_max}$ of string
DC Disconnect Switch	Allows safe isolation of panels from inverter for maintenance	Mandatory per most codes; rated $\geq 1.25 \times I_{sc}$
AC Breaker / MCB	Protects AC output wiring from inverter to distribution board	Sized to inverter rated output current $\times 1.25$
Bidirectional Energy Meter	Measures import and export to grid; required for FiT payments	Supplied/approved by utility; smart meter with comms preferred
Surge Protection Device (SPD)	Clamps voltage spikes from lightning or grid switching on DC & AC sides	Type 2 SPD on DC string; Type 2 on AC output; Type 1 at main panel
Mounting Racking	Supports panels at correct tilt and azimuth; transfers wind/snow loads to structure	Designed to local wind speed zone; min 4 clamps per panel
DC Cabling	Single-core, UV-resistant, double-insulated solar cable	4 mm ² typical residential; sized to keep I^2R loss $< 1\text{--}2\%$ per run

What comes next — SOLAR-004: Solar Panels in Depth — understanding I-V curves, temperature coefficients, shading losses, panel datasheets, and how to select the right module for your project.